25900361 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT203 - AI AND MACHINE LEARNING FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Describe business growth prediction using ML. [U] / [CO1]

   a). Chalk b). ML algorithms

   c). Pen d). Bench

2. State what Python is best known for. [K] / [CO2]

   a). Simplicity & readability b). Complex syntax

   c). Hardware design d). Game consoles

3. Identify a real-life example of supervised learning [K] / [CO3]

   a). Movie recommendation system

   b). Predicting house prices from past data

   c). Clustering customers by buying habits

   d). Robot learning to walk by trial and error

4. Recognize the role of activation function in forward propagation. [K] / [CO4]

a). Error correction b). Non-linear transformation

c). Gradient update d). Weight storage

5. Describe the main goal of AI. [U] / [CO1]

a). Mimic human intelligence b). Replace hardware

c). Enhance internet speed d). Design robots only

6. Classify Machine Learning as a subset of. [U] / [CO1]

a). Artificial Intelligence b). Physics

c). Chemistry d). Civil Engineering

7. State what Pandas is mainly used for. [K] / [CO2]

a). Data manipulation b). Graphics

c). Networking d). Games

8. Describe the goal of linear regression. [U] / [CO3]

a). Minimize the sum of absolute errors b). Minimize the sum of squared errors

c). Maximize the correlation coefficient d). Minimize the number of features

9. Summarize the working of perceptron. [U] / [CO4]

a). Weighted sum + Activation b). Random guessing

c). Direct storage d). Data memorization

10. Compare sigmoid and step function. [U] / [CO4]

a). Continuous vs Discrete b). Fast vs Slow

c). Biological vs Artificial d). Strong vs Weak

11. List common uses of Pandas. [K] / [CO2]

a). Series and DataFrame b). Array and list

c). Graph and chart d). Tuple and dict

12. List the main features Seaborn provides. [K] / [CO2]

a). High-level API for visuals b). Only text

c). Only tables d). Only arrays

13. Classify hierarchical clustering as a type of machine learning technique. [U] / [CO3]

a). Supervised learning b). Unsupervised learning

c). Semi-supervised learning d). Reinforcement learning

14. Explain what correlation analysis shows. [U] / [CO2]

a). Cooking steps b). Relationship between variables

c). Car speed d). Time

15. Differentiate forward and backpropagation. [U] / [CO4]

a). Both same b). Forward: learning, Backward: prediction

c). Forward: prediction, Backward: learning d). None

16. Use the Pandas function that shows the first rows of data. [A] / [CO2]

a). tail() b). head()

c). top() d). first()

17. Identify how outliers can be handled. [K] / [CO2]

a). Removing or transforming b). Sorting

c). Copying d). Joining

18. Select the architecture type of FFNN. [U] / [CO4]

a). Random b). Cyclic

c). Hierarchical tree d). Layered structure

19. Define a biological neuron. [K] / [CO4]

a). A nerve cell b). A processing unit in AI

c). A decision tree d). A simple circuit

20. State which kernel is the default in Jupyter. [K] / [CO2]

a). C++ b). Java

c). C d). Python

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Compare KNN with decision tree algorithms in terms of training and prediction time. [U] / [CO3]

22. Define feature engineering and its role in ML. [K] / [CO2]

23. Explain the basic components of a reinforcement learning system [U] / [CO3]

24. Define a neural network and state its importance in machine learning. [K] / [CO4]

25. Discuss the importance of normalization and standardization. [U] / [CO2]

26. Illustrate the steps to configure a virtual environment in PyCharm. [A] / [CO2]

27. Describe one application of supervised learning. [U] / [CO3]

28. Differentiate between high learning rate and low learning rate effects on convergence. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Discuss the gradient descent optimization algorithm in detail. Differentiate between batch, stochastic, and mini-batch gradient descent. [U] / [CO4]

30. Explain the step-by-step procedure of the K-Means Clustering algorithm with an example dataset. [U] / [CO3]

31. Differentiate between Artificial Intelligence, Machine Learning, and Deep Learning with suitable examples. [U] / [CO1]

32. Describe the importance of data visualization in machine learning. [U] / [CO2]

[10x2 = 20]